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Start-Stop Duty Cycle Sizing Worksheet

For OEM designers and maintenance engineers sizing motors for indexing, shuttling and other frequent start-stop duty

Work through the steps with real logged data rather than estimates. Steps 1 to 4 decide whether a motor can carry the cycle thermally; Steps 5 and 6 protect brakes, brushes and gearboxes. Use SI units throughout: kg·m², rad/s, Nm, A and seconds.

Step 1 — Capture the real duty cycle

- [ ] Log armature current and speed over at least ten consecutive cycles in normal production.
- [ ] Record acceleration, running, braking and rest times for a typical cycle, and the total cycle time.
- [ ] Record starts per hour at peak production, not the shift average.
- [ ] Note any reversals, and whether the load is ever started against a jam or a full hopper.
- [ ] Measure voltage at the motor terminals during acceleration.

Step 2 — Inertia and acceleration

- [ ] List every load inertia at the output: tables, rollers, drums and carried product.
- [ ] Reflect to the motor shaft: load inertia / (gear ratio)²; add motor and gearbox input inertia.
- [ ] Keep the external-to-motor inertia ratio below about 5:1 for quick, controlled starts.
- [ ] Acceleration time = total inertia × speed in rad/s / (motor torque – load torque).
- [ ] Confirm the gearbox rating covers the peak acceleration and braking torque.

Step 3 — Heat per start and stop

- [ ] Kinetic energy at speed = 0.5 × J × (speed in rad/s)².
- [ ] Direct-on-line start: armature heat is about equal to the kinetic energy at no-load speed, more under load.
- [ ] Current-limited start: armature heat = (current limit)² × armature resistance × acceleration time.
- [ ] Plug stop dissipates about 3 × the kinetic energy; plug reversal about 4 ×.
- [ ] Average heating in watts = energy per event (J) × events per hour / 3,600.

Step 4 — RMS current and thermal check

- [ ] RMS current = square root of [sum of (current² × time) / cycle time], including braking current.
- [ ] For self-ventilated motors count only part of the rest time; half is a cautious allowance.
- [ ] Compare RMS current with the continuous rated current, keeping margin for 45–50 °C ambients and supply sag.
- [ ] Repeat the check at the highest start rate production may ever request.
- [ ] If the margin is thin, reduce the current limit, the inertia or the start rate before upsizing the frame.

Step 5 — Braking and holding

- [ ] Use regenerative or dynamic braking for the stopping duty.
- [ ] Dynamic braking resistor: total resistance = back-EMF / maximum braking current; motor share = armature resistance / total resistance.

- [ ] On rectified mains supplies, regenerative braking needs a brake chopper and resistor.
- [ ] Engage the fail-safe brake at or near zero speed; keep its friction energy for emergency stops.
- [ ] Check brake release and engagement times, and permissible energy per operation and per hour, against the brake data.

Step 6 — Brushes, gearbox and technology choice

- [ ] Set the starting current limit within the motor's peak current rating.
- [ ] Never plug-reverse a ferrite PMDC motor at full voltage; brake to near zero speed first.
- [ ] Track brush wear against cycle count as well as running hours.
- [ ] Use soft ramps to limit gear shock and backlash impact at reversal.
- [ ] Consider a BLDC motor where brush wear, rotor heating or regenerative braking needs limit a brushed design.

Starting and Braking Methods Compared

Method	Peak motor current	Heat left in the motor	Use when
Direct-on-line start	Stall current (V / armature resistance)	About the kinetic energy at no-load speed, per start	Rare starts, small inertia
Current-limited ramp start	Set limit, within peak rating	About 2 × limit / stall current of the DOL figure	Frequent starts (default)
Coast to stop	None	None	Long cycles, loose stopping accuracy
Dynamic braking resistor	Back-EMF / total resistance	Armature share of the kinetic energy	Simple controlled stops
Regenerative braking	Set by controller	Only I^2R during braking	Batteries, high cycle rates
Plug stop or reversal	Up to about 2 × stall	3 × (stop) or 4 × (reversal) kinetic energy	Avoid on PMDC motors
Brake used as stopping device	None	None in motor; all energy into the lining	Emergency stops only

Energy relationships assume a pure inertia load on a PMDC motor; load torque increases the heat per start. Confirm peak current, duty and brake ratings with the manufacturer.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.